

Day 13 Plan

Maths: Mensuration 3D · Reasoning: Seating Arrangement · Science: Biology 1:
Cell, Organelles, Tissues & Classification

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 12 recall	15 min	10 questions from Day 12, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	8 lessons with quick checks (≈65 min) → 20-question practice set → Testbook PYQs: cell, organelles, tissues (S tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Sports & awards (static) · also check your city intimation slip today — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 12 recall (answers on the last page)

#	Question	Your answer
1	General formula of alkanes?	
2	Vinegar contains which acid?	
3	Catalyst for hydrogenation of oils?	
4	Main component of CNG?	
5	Hottest zone of a candle flame?	
6	Common name of N_2O ?	
7	Teflon is used for?	
8	Fuel with the highest calorific value?	
9	Empirical relation for mode?	
10	Area of an equilateral triangle of side 6?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Mensuration 3D

Core idea

Every 3D question is one of three things: a **volume** (capacity, melting, recasting), a **surface area** (painting, wrapping, cost), or a **ratio** (radius doubled, etc.). When a solid is melted and recast, the **volume stays the same**. Use $\pi = 22/7$ when the radius is a multiple of 7.

Formulas

Concept	Formula / rule
Cube (side a)	Volume a^3 · total surface $6a^2$ · lateral $4a^2$ · diagonal $a\sqrt{3}$
Cuboid	Volume lbh · total surface $2(lb + bh + hl)$ · diagonal $\sqrt{l^2 + b^2 + h^2}$
Cylinder	Volume $\pi r^2 h$ · curved $2\pi rh$ · total $2\pi r(r + h)$
Cone	Volume $\frac{1}{3}\pi r^2 h$ · slant $l = \sqrt{r^2 + h^2}$ · curved πrl · total $\pi r(l + r)$
Sphere	Volume $\frac{4}{3}\pi r^3$ · surface $4\pi r^2$
Hemisphere	Volume $\frac{2}{3}\pi r^3$ · curved $2\pi r^2$ · total $3\pi r^2$
Units	$1 \text{ m}^3 = 1000 \text{ litres}$ · $1 \text{ litre} = 1000 \text{ cm}^3$
Scaling	Lengths $\times k \rightarrow$ areas $\times k^2 \rightarrow$ volumes $\times k^3$

Patterns — the method and one solved example each

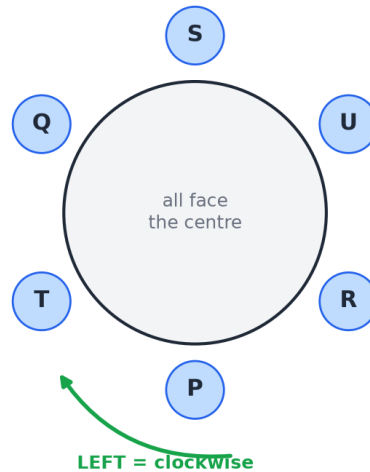
Pattern	Method	Solved example
1. Cube	$a^3 \rightarrow a$	Volume 343 $\rightarrow$ side 7 $\rightarrow$ total surface $6 \times 49 = \mathbf{294}$
2. Cuboid diagonal	$\sqrt{l^2 + b^2 + h^2}$	$12 \times 4 \times 3 \rightarrow \sqrt{169} = \mathbf{13}$; volume 144
3. Cylinder	$\pi r^2 h, 2\pi rh$	$r = 7, h = 10 \rightarrow$ volume 1540 , curved surface 440
4. Cone	Slant first	$r = 6, h = 8 \rightarrow l = 10 \rightarrow$ curved surface $60\pi \approx \mathbf{188.6}$; volume $96\pi \approx \mathbf{301.7}$
5. Sphere	$\frac{4}{3}\pi r^3$	$r = 21 \rightarrow$ volume 38,808 ; surface 5,544
6. Recasting count	Volume ratio	Sphere $r = 6$ into spheres of $r = 2 \rightarrow (6/2)^3 = \mathbf{27}$
7. Sphere $\rightarrow$ cylinder	Equal volumes	Sphere $r = 3$ recast as a cylinder of $r = 3 \rightarrow \frac{4}{3} \times 27 = 9h \rightarrow h = \mathbf{4 \text{ cm}}$
8. Capacity	$\text{m}^3 \rightarrow \text{litres}$	Tank $2 \times 1.5 \times 1 \text{ m} = 3 \text{ m}^3 = \mathbf{3000 \text{ L}}$
9. Scaling	k^2, k^3	Sphere's radius doubled $\rightarrow$ surface $\times \mathbf{4}$, volume $\times \mathbf{8}$
10. Hemisphere	$3\pi r^2$	$r = 7 \rightarrow$ total surface $3 \times 22/7 \times 49 = \mathbf{462}$

Traps

- Curved vs total surface — an open tank or bowl has no top: add only the faces that are actually covered.
- Cone: use the **slant** height for surface area, the **vertical** height for volume.
- A cone has $\frac{1}{3}$ the volume of a cylinder with the same base and height.

B · REASONING — Seating Arrangement

Linear row facing north: their left = your left as you draw it. **Facing south:** left and right are reversed. **Circle facing the centre:** left = **clockwise**, right = anticlockwise. Facing outward: reversed.



Worked example — circular arrangement (the figure above)

Six people P, Q, R, S, T, U sit around a circle facing the centre. Q is second to the left of P. R is immediately to the right of P. S is opposite P. T sits between P and Q.

Left = clockwise: $P \rightarrow T \rightarrow Q$. R is one seat anticlockwise from P. S is opposite P. The last seat goes to U.

Opposite Q is **R** · immediately to the left of S is **U**

Worked example — linear row

A, B, C, D, E sit in a row facing north. C is in the middle. A is immediately to the left of C. B is at the right end. D is not next to A.

Seats 1-5 left to right: $C = 3, A = 2, B = 5 \rightarrow D$ can't be 1 (next to A) $\rightarrow D = 4, E = 1$.

E A C D B

Patterns, with solved examples

Pattern	Example	Solution
Linear end	In the row above, who sits at the left end?	E
Between	In a row of 7 facing north, A is 3rd from the left and B is 2nd from the right. How many sit between them?	Seats 3 and 6 $\rightarrow$ 2
Circle opposite	In the circle above, who is opposite Q?	R
Circle neighbours	In the circle above, who are T's neighbours?	P and Q
Facing south	Five people face south. X is at the extreme left of the row as they see it. On your drawing (north up) X is at which end?	Facing south, their left is east $\rightarrow$ X is at the east (right-hand) end of your drawing

Speed rule

Draw the seats before reading the clues a second time. Fix the definite clues, then try the two or three possible cases for the rest.

C · SCIENCE — Biology 1: Cell, Organelles, Tissues & Classification

8 lessons. Cell, organelles and tissues were 18 of 210 science questions in the 2026 papers (R) and were flagged in your own audit (Y). Lesson 4 (organelles) is the most-asked table in biology.

Lesson 1 · Discovery of the cell and cell theory (about 6 min)

Scientist	Contribution
Robert Hooke (1665)	Saw cells in cork (dead) and coined the word cell
Leeuwenhoek (1674)	First to see living cells (bacteria, protozoa) with his own microscope
Robert Brown (1831)	Discovered the nucleus
Purkinje (1839)	Coined ' protoplasm '
Schleiden & Schwann (1838–39)	Cell theory : all plants and animals are made of cells
Virchow (1855)	All cells arise from pre-existing cells

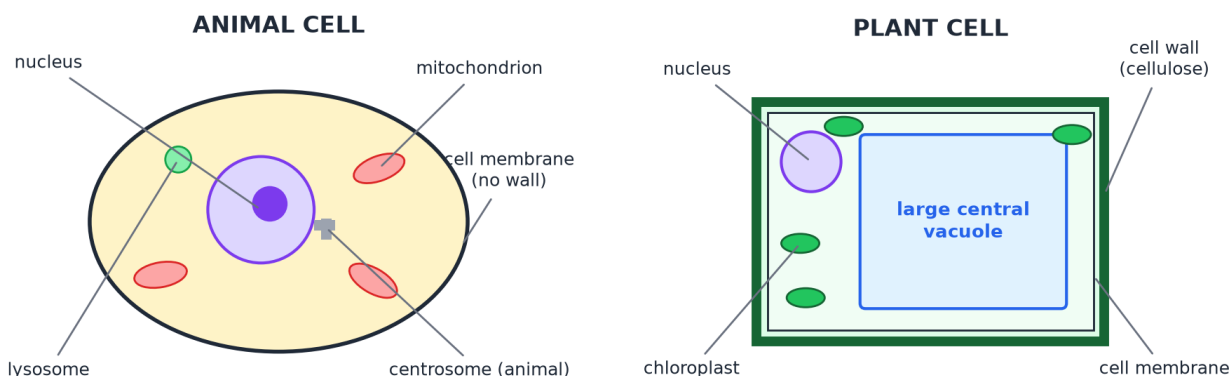
Smallest cell: **Mycoplasma (PPLO)**. Largest single cell: **ostrich egg**. Longest cell in the human body: **nerve cell (neuron)**. Unicellular organisms: Amoeba, Paramecium, Chlamydomonas, bacteria.

Quick check 1 — answer out loud before moving on (answers at the end)

a) Who coined the word 'cell'? b) Longest cell in the human body?

Lesson 2 · Prokaryotes vs eukaryotes; plant vs animal cells (about 8 min)

	Prokaryotic	Eukaryotic
Nucleus	No membrane — DNA lies in a nucleoid	True nucleus with a membrane
Membrane-bound organelles	Absent	Present (mitochondria, Golgi, ER ...)
Examples	Bacteria, blue-green algae (cyanobacteria)	Plants, animals, fungi, protists



Feature	Plant cell	Animal cell
Cell wall	Present (cellulose)	Absent
Plastids (chloroplasts)	Present	Absent
Vacuole	One large central vacuole	Small or absent
Centrosome / centrioles	Usually absent	Present
Shape	Fixed, often rectangular	Irregular, rounded

Fungal cell walls are made of **chitin**; bacterial walls of peptidoglycan.

Quick check 2 — answer out loud before moving on (answers at the end)

Name two structures found in plant cells but not in animal cells.

Lesson 3 • Membrane, osmosis and the nucleus (about 8 min)

The **plasma membrane** (lipids + proteins) is **selectively permeable**. Substances cross by **diffusion** (high → low concentration). **Osmosis** is the movement of **water** through a selectively permeable membrane from a dilute to a concentrated solution.

Outside solution	Water moves	Cell
Hypotonic (more dilute)	Into the cell	Swells; an animal cell (RBC) may burst; a plant cell becomes turgid
Isotonic	No net movement	Unchanged
Hypertonic (more concentrated)	Out of the cell	Shrinks; in plant cells the contents pull away from the wall = plasmolysis

Raisins swell in water and pickles are preserved in salt by osmosis. Amoeba takes in food by **endocytosis**. The **nucleus** controls the cell; it holds **chromosomes** (DNA + protein) and the **nucleolus** (makes ribosomes).

Quick check 3 — answer out loud before moving on (answers at the end)

a) Movement of water across a selectively permeable membrane is called? b) What happens to a red blood cell in a hypotonic solution?

Lesson 4 • Cell organelles (about 10 min)

Organelle	Job	Remember
Mitochondria	Aerobic respiration; make ATP	Powerhouse of the cell ; have their own DNA and ribosomes
Plastids	Chloroplast — photosynthesis · chromoplast — colour · leucoplast — stores starch, oil, protein	Only in plants; own DNA
Ribosomes	Make proteins	No membrane; free or on rough ER
Endoplasmic reticulum	Rough ER — proteins · smooth ER — lipids; detoxifies drugs and poisons (liver)	Transport network
Golgi apparatus	Packages, stores and secretes; forms lysosomes	Named after Camillo Golgi
Lysosomes	Digest waste and worn-out parts	Suicide bags — burst and digest their own cell
Vacuoles	Storage; keep plant cells turgid	Large in plants · contractile vacuole in Amoeba removes extra water
Centrosome	Helps in cell division	Animal cells

Quick check 4 — answer out loud before moving on (answers at the end)

a) Which organelle is called the 'suicide bag'? b) Which two organelles have their own DNA? c) Where are proteins made?

Lesson 5 • Cell division and chromosomes (about 7 min)

	Mitosis	Meiosis
Where	Body (somatic) cells	Reproductive organs (to make gametes)
Daughter cells	2, identical to the parent	4, different
Chromosome number	Same as parent (46 in humans)	Halved (23 in human sperm and egg)
Purpose	Growth and repair	Sexual reproduction; variation

Humans have **46 chromosomes (23 pairs)**. DNA's double-helix structure was described by **Watson and Crick (1953)**. A **gene** is a section of DNA that carries one instruction.

Quick check 5 — answer out loud before moving on (answers at the end)

a) Number of chromosomes in a human sperm? b) Which division forms gametes?

Lesson 6 • Plant tissues (about 9 min)

Tissue	Type	Job / where
Meristem	Dividing	Apical — growth in length (tips) · lateral (cambium) — growth in girth · intercalary — at nodes, leaf bases
Parenchyma	Simple, living	Storage; with chloroplasts = chlorenchyma; with air spaces = aerenchyma (helps aquatic plants float)
Collenchyma	Simple, living	Flexibility (leaf stalks); walls thickened at the corners
Sclerenchyma	Simple, dead	Strength; walls thickened with lignin ; coconut husk
Xylem	Complex	Carries water and minerals upward (tracheids, vessels — mostly dead)
Phloem	Complex	Carries food both ways (sieve tubes, companion cells — mostly living)
Epidermis · cork	Protective	Stomata with guard cells · cork cells have suberin (waterproof)

Quick check 6 — answer out loud before moving on (answers at the end)

a) Which tissue carries food in plants? b) Which tissue gives flexibility to plants?

Lesson 7 • Animal tissues (about 9 min)

Tissue	Kinds and examples
Epithelial (covering)	Squamous — skin surface, lining of mouth · cuboidal — kidney tubules, glands · columnar — intestine lining · ciliated — windpipe
Connective	Blood (fluid matrix = plasma) · bone (calcium, phosphorus) · cartilage (nose, ear, trachea rings) · tendon: muscle to bone · ligament: bone to bone · areolar (under skin) · adipose (fat, insulation)
Muscular	Striated/voluntary/skeletal — on bones · smooth/involuntary — gut, iris, blood vessels · cardiac — striated but involuntary , branched, never tires
Nervous	Neuron: dendrites → cell body → axon; carries impulses

Quick check 7 — answer out loud before moving on (answers at the end)

a) What connects muscle to bone? b) Which muscle is striated but involuntary?

Lesson 8 • Classification basics (quick) (about 6 min)

Kingdom (Whittaker, 1969)	Features	Examples
Monera	Prokaryotic	Bacteria, blue-green algae
Protista	Unicellular eukaryotes	Amoeba, Paramecium, Euglena, diatoms
Fungi	Heterotrophic, chitin cell wall	Yeast, mushroom, Penicillium
Plantae	Multicellular, autotrophic, cellulose wall	Mosses, ferns, flowering plants
Animalia	Multicellular, heterotrophic, no cell wall	Sponges to mammals

Carolus Linnaeus: binomial nomenclature, father of taxonomy — Homo sapiens (genus capitalised, species lower-case, italics). Order of ranks: Kingdom → Phylum → Class → Order → Family → Genus → Species. Viruses are not placed in the five kingdoms.

Classic traps: **whale and bat are mammals**; penguin is a bird; **starfish and jellyfish are not fish**; silverfish is an insect.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Who proposed the five-kingdom classification? b) Is a whale a fish?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Discovery	Hooke cell (cork) · Leeuwenhoek living cells · Brown nucleus · Schleiden & Schwann cell theory · Virchow
Records	Smallest Mycoplasma · largest ostrich egg · longest human cell neuron
Plant vs animal	Plant: cell wall, plastids, big vacuole · animal: centrosome · fungi wall chitin
Osmosis	Water dilute → concentrated · hypotonic swells · hypertonic shrinks (plasmolysis)
Organelles	Mitochondria powerhouse (ATP) · lysosome suicide bag · ribosome protein · Golgi packaging · mitochondria + plastids own DNA
Division	Mitosis 2 identical · meiosis 4, half chromosomes · humans 46, gametes 23 · DNA Watson & Crick
Plant tissues	Xylem water up · phloem food · collenchyma flexibility · sclerenchyma lignin · cambium girth
Animal tissues	Tendon muscle-bone · ligament bone-bone · cardiac striated + involuntary
Classification	Whittaker 5 kingdoms · Linnaeus binomial · whale, bat mammals · starfish not fish

Science practice set — 20 questions · target 6 minutes

#	Question
1	The term 'cell' was coined by: (a) Leeuwenhoek (b) Robert Brown (c) Schwann (d) Robert Hooke
2	The nucleus was discovered by: (a) Robert Hooke (b) Virchow (c) Robert Brown (d) Purkinje
3	The longest cell in the human body is the: (a) muscle cell (b) nerve cell (c) red blood cell (d) bone cell
4	Prokaryotic cells lack: (a) a membrane-bound nucleus (b) DNA (c) ribosomes (d) a cell membrane
5	Which is found in plant cells but not in animal cells? (a) Mitochondria (b) Ribosomes (c) Nucleus (d) Cell wall
6	The cell wall of fungi is made of: (a) chitin (b) cellulose (c) lignin (d) protein
7	Movement of water through a selectively permeable membrane is: (a) diffusion (b) transpiration (c) endocytosis (d) osmosis
8	A plant cell placed in a strong salt solution shows: (a) turgidity (b) plasmolysis (c) bursting (d) no change
9	The powerhouse of the cell is the: (a) ribosome (b) Golgi apparatus (c) mitochondrion (d) nucleus
10	The 'suicide bags' of the cell are: (a) lysosomes (b) vacuoles (c) plastids (d) ribosomes
11	Proteins are made by: (a) lysosomes (b) centrosomes (c) ribosomes (d) vacuoles
12	Which organelle has its own DNA? (a) Chloroplast (b) Golgi apparatus (c) Lysosome (d) Ribosome
13	Gametes are formed by: (a) mitosis (b) binary fission (c) budding (d) meiosis
14	The number of chromosomes in a human egg cell is: (a) 46 (b) 22 (c) 23 (d) 44
15	Water and minerals are carried in plants by: (a) xylem (b) phloem (c) cambium (d) cork
16	The tissue responsible for growth in girth (thickness) is: (a) apical meristem (b) lateral meristem (cambium) (c) collenchyma (d) phloem
17	Muscle is joined to bone by a: (a) ligament (b) cartilage (c) tendon (d) nerve
18	Cardiac muscle is: (a) smooth and voluntary (b) striated and voluntary (c) striated and involuntary (d) smooth and involuntary
19	The five-kingdom classification was proposed by: (a) Linnaeus (b) R. H. Whittaker (c) Darwin (d) Aristotle
20	Which of these is a mammal? (a) Penguin (b) Starfish (c) Shark (d) Bat

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	d	Lesson 1
2	c	Lesson 1
3	b	Lesson 1
4	a	Lesson 2
5	d	Lesson 2
6	a	Lesson 2
7	d	Lesson 3
8	b	Lesson 3
9	c	Lesson 4
10	a	Lesson 4
11	c	Lesson 4
12	a	Lesson 4
13	d	Lesson 5
14	c	Lesson 5
15	a	Lesson 6
16	b	Lesson 6
17	c	Lesson 7
18	c	Lesson 7
19	b	Lesson 8
20	d	Lesson 8

Quick checks

QC	Answers
1	a) Robert Hooke b) The nerve cell (neuron)
2	Any two: cell wall, plastids (chloroplasts), large central vacuole
3	a) Osmosis b) Water enters; it swells and may burst
4	a) Lysosome b) Mitochondria and plastids (chloroplasts) c) Ribosomes
5	a) 23 b) Meiosis
6	a) Phloem b) Collenchyma
7	a) Tendon b) Cardiac muscle
8	a) R. H. Whittaker b) No — it is a mammal

Day 12 recall (Block 0)

#	Answer	#	Answer
1	C_nH_{2n+2}	6	Laughing gas
2	Acetic (ethanoic) acid, 5-8%	7	Non-stick cookware
3	Nickel	8	Hydrogen
4	Methane	9	Mode = 3 median – 2 mean
5	Outermost zone	10	$9\sqrt{3} \approx 15.59$

Send me tonight

Day 12 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.